

5 Monocotyledons are herbaceous flowering plants that differ from dicotyledons in the distribution of tissues in their roots, stems and leaves.

(a) In monocotyledons and dicotyledons, water and mineral ions are taken up from the soil solution and move to the xylem by taking the apoplast and symplast pathways.

(i) State the term used to describe:

- the passive movement of substances, such as mineral ions, across the cell surface membrane into the root cell using membrane transport proteins

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- the process by which water passes across the cell surface membrane from the soil solution into a root cell.

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[2]

(ii) Water that has entered a root cell can move passively through plasmodesmata to an adjacent cell.

Explain what causes water to move from a root cell to an adjacent cell.

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..... [1]

(b) Figure 5.1 is a transverse section (TS) through the root of a monocotyledon.

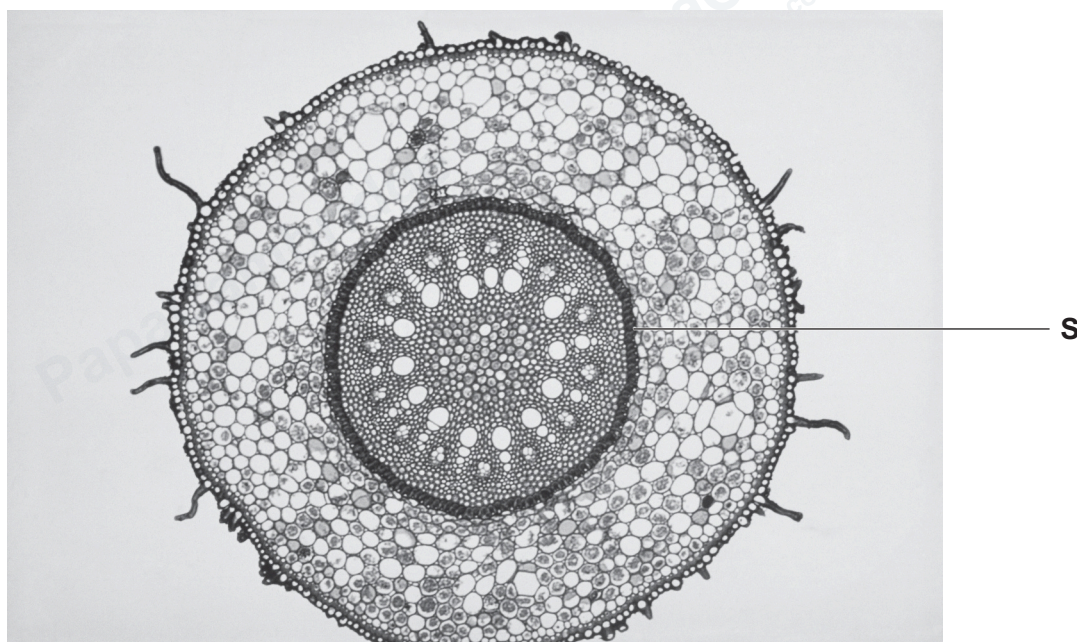


Figure 5.1



- (i) The monocotyledon root shown in Figure 5.1 has been stained to show the presence of a layer of cells, labelled **S**, where suberin is located. This layer of cells is also present in dicotyledons.

State:

- the name of the layer of cells in which suberin is located
- the role of suberin in this layer.

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..... [2]

- (ii) Describe the similarities **and** differences in the distribution of tissues between the TS of the root of the herbaceous monocotyledon shown in Figure 5.1 and a TS of the root of a herbaceous dicotyledon.

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[Total: 9]